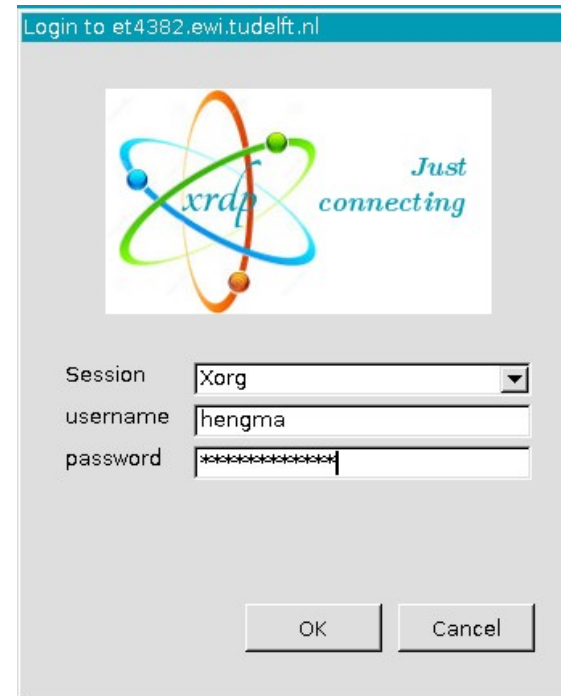
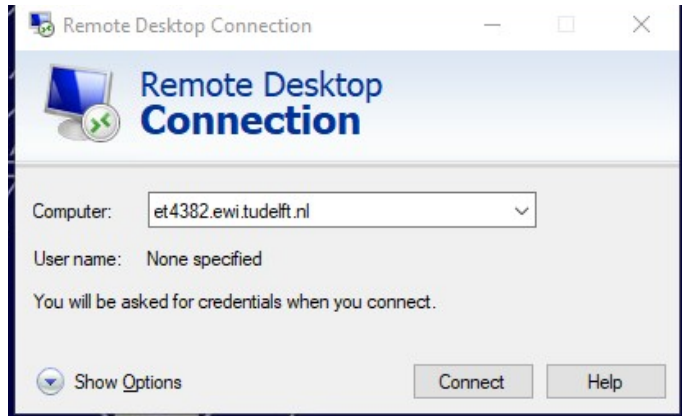


Content

- Connecting to server
- Candence introduction
- Creat a LC filter
- Run simulations
- Plotting waveforms
- Performing FFT
- Parametric analysis
- Reference

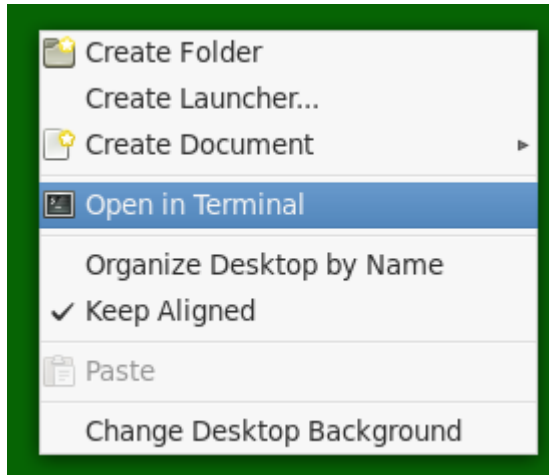
Connect to server

- 1. connect to TU network (weblogin.tudelft.nl)
- 2. use remote desktop connection and type in “et4382.ewi.tudelft.nl”. Then click connect
- 3. type in you netid and netid password



Set your environment

First, you set up your working directory where you will use Cadence for the entire semester. You only need to do this once. (Hit ENTER after each line.)



```
cd ~  
mkdir tsmcBCD  
cd ~/tsmcBCD  
source /opt/ei/DK/tsmc/oa180/mini018BCDG2/216A/et4382/start
```

To start Cadence, type the following.

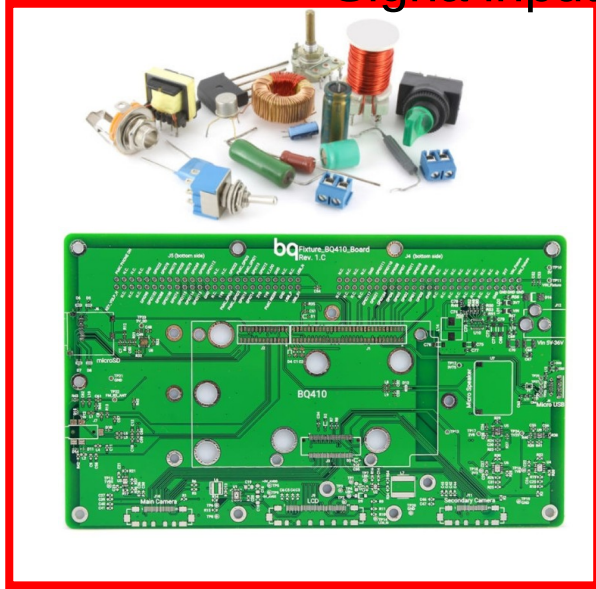
```
cd ~/tsmcBCD  
source sourceme  
virtuoso&
```

You need to do this every time you start Cadence. After Cadence starts, you should see the CIW window:

How to use cadence?



↓
Signal input



↓
Signal output



1. Where is the source?
2. Where is the device?
3. Where is the board/schematic?

Cadence

■ Source model



■ Device model

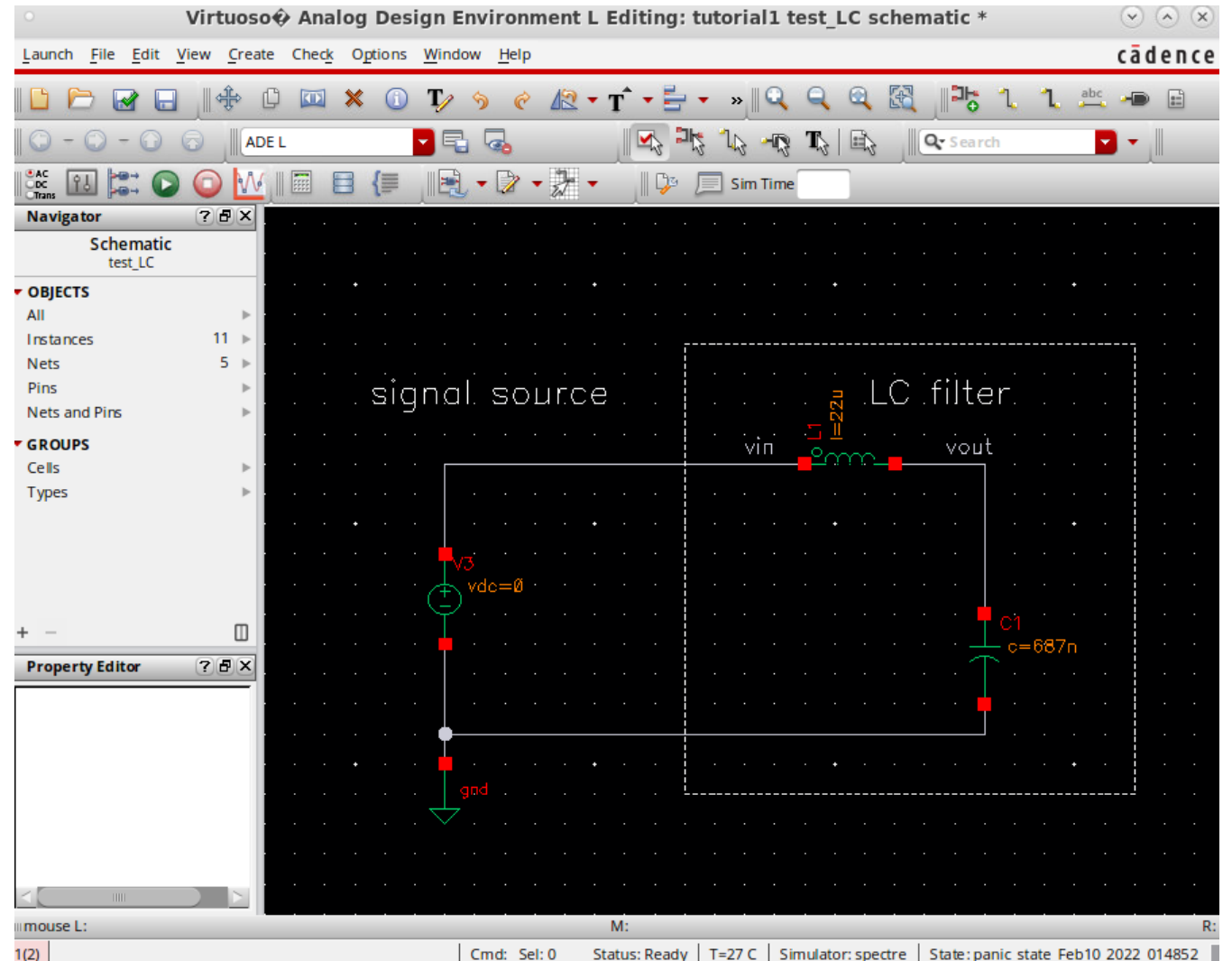
■ Schematic



■ Readout simulation results

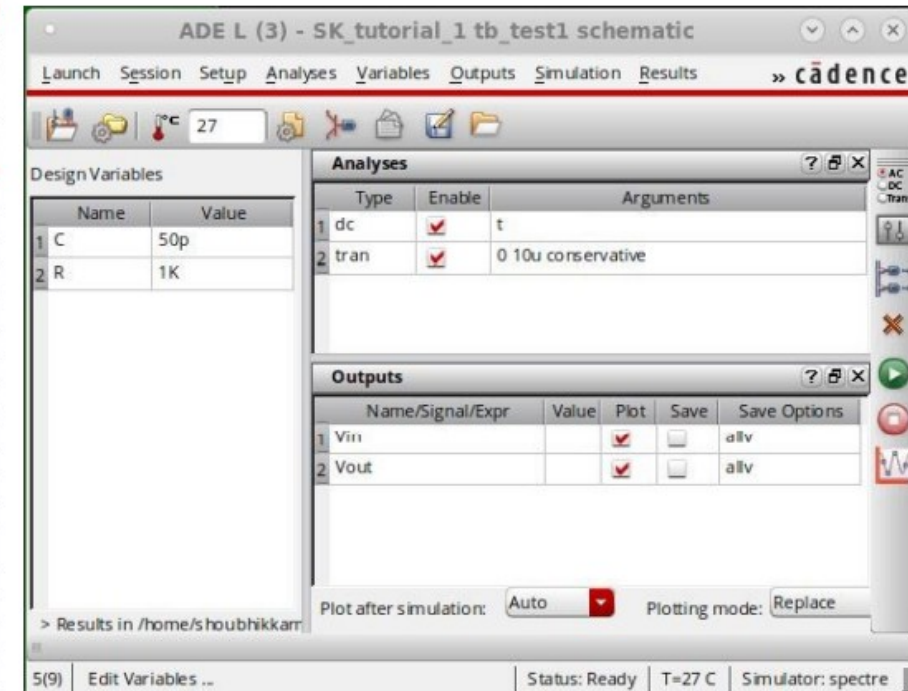
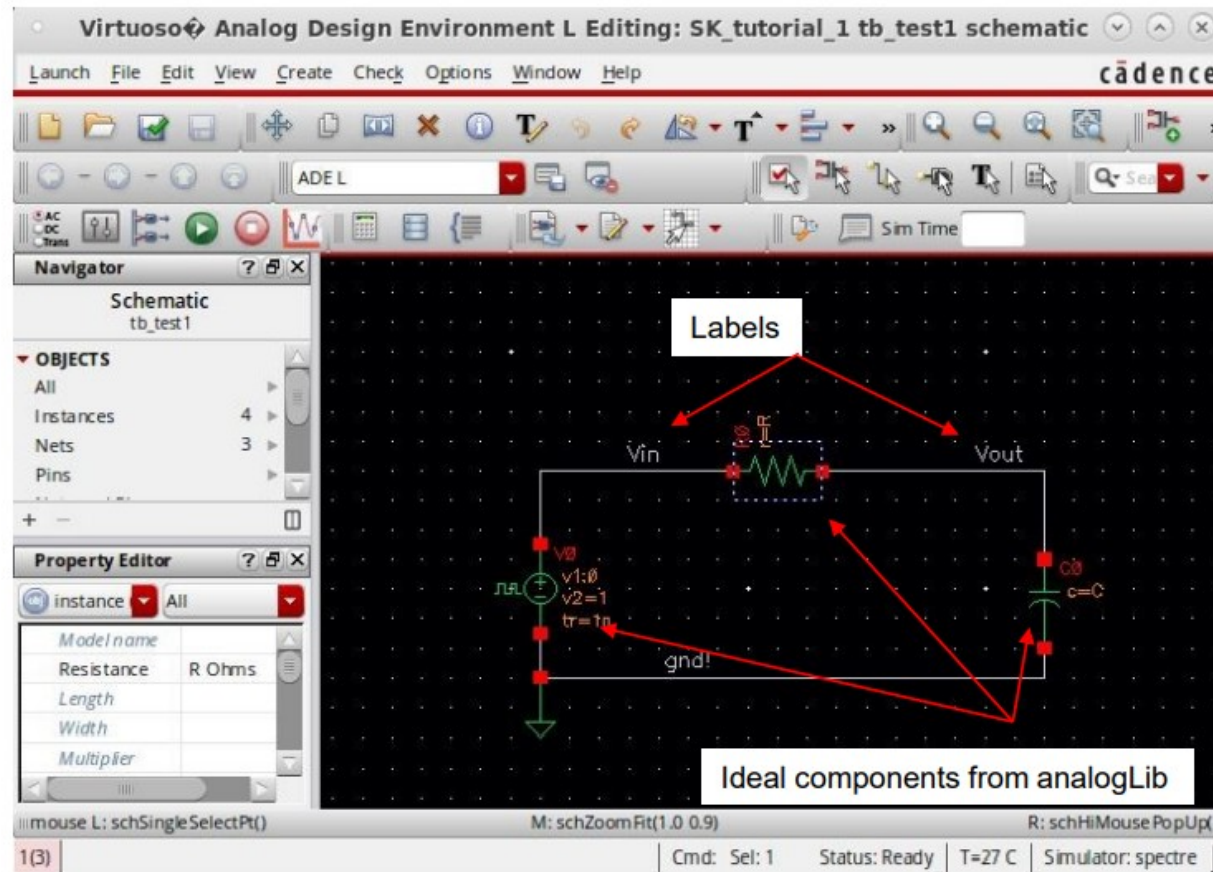
Create a schematic

1. Create a empty schematic
2. Add devices
3. Add sources
4. Add lable



Running a simulation

- Once your testbench is set up you need to launch the simulator. From the Schematic editor itself: Launch> ADE L. You will be asked some questions about licenses. Simply keep answering yes.



Analysis options

Run simulation.
(Check and save first)

- It is always useful to run a DC analysis and check operating points (Especially for linear/ non-periodically switching circuits). To do this enable DC analysis from analysis options and select "Save DC op point." After running the simulation, Results> Annotate > DC node voltages. The DC node voltages should show up in the schematic.
- Component parameters can be given variables in case of sweeps, etc. To load the variable into ADE (R,C in this case) variables>Copy from cellview.

DC/AC/Transient simulation

Choosing Analyses -- ADE L (1)

Analysis

☐ tran	☒ dc	☐ ac	☐ noise
☐ xf	☐ sens	☐ dcmatch	☐ acmatch
☐ stb	☐ pz	☐ lf	☐ sp
☐ envlp	☐ pss	☐ pac	☐ pstb
☐ pnoise	☐ pxf	☐ psp	☐ qpss
☐ qpac	☐ qpnoise	☐ qpxf	☐ qpssp
☐ hb	☐ hbac	☐ hbstb	☐ hbnoise
☐ hbsp	☐ hbxf		

DC Analysis

Save DC Operating Point ☒

Hysteresis Sweep ☐

Sweep Variable

☐ Temperature

☐ Design Variable

☐ Component Parameter

☐ Model Parameter

Enabled ☒

Options...

OK Cancel Defaults Apply Help

Choosing Analyses -- ADE L (1)

Analysis

☐ tran	☐ dc	☒ ac	☐ noise
☐ xf	☐ sens	☐ dcmatch	☐ acmatch
☐ stb	☐ pz	☐ lf	☐ sp
☐ envlp	☐ pss	☐ pac	☐ pstb
☐ pnoise	☐ pxf	☐ psp	☐ qpss
☐ qpac	☐ qpnoise	☐ qpxf	☐ qpssp
☐ hb	☐ hbac	☐ hbstb	☐ hbnoise
☐ hbsp	☐ hbxf		

AC Analysis

Sweep Variable

☒ Frequency

☐ Design Variable

☐ Temperature

☐ Component Parameter

☐ Model Parameter

☐ None

Sweep Range

☒ Start-Stop

Start Stop

☐ Center-Span

Sweep Type

☒ Points Per Decade

☐ Number of Steps

Add Specific Points ☐

Specialized Analyses

Enabled ☒

Options...

OK Cancel Defaults Apply Help

Choosing Analyses -- ADE L (1)

Analysis

☒ tran	☐ dc	☐ ac	☐ noise
☐ xf	☐ sens	☐ dcmatch	☐ acmatch
☐ stb	☐ pz	☐ lf	☐ sp
☐ envlp	☐ pss	☐ pac	☐ pstb
☐ pnoise	☐ pxf	☐ psp	☐ qpss
☐ qpac	☐ qpnoise	☐ qpxf	☐ qpssp
☐ hb	☐ hbac	☐ hbstb	☐ hbnoise
☐ hbsp	☐ hbxf		

Transient Analysis

Stop Time

Accuracy Defaults (errpreset)

☒ conservative ☐ moderate ☐ liberal

☐ Transient Noise

Dynamic Parameter ☐

Enabled ☒

Options...

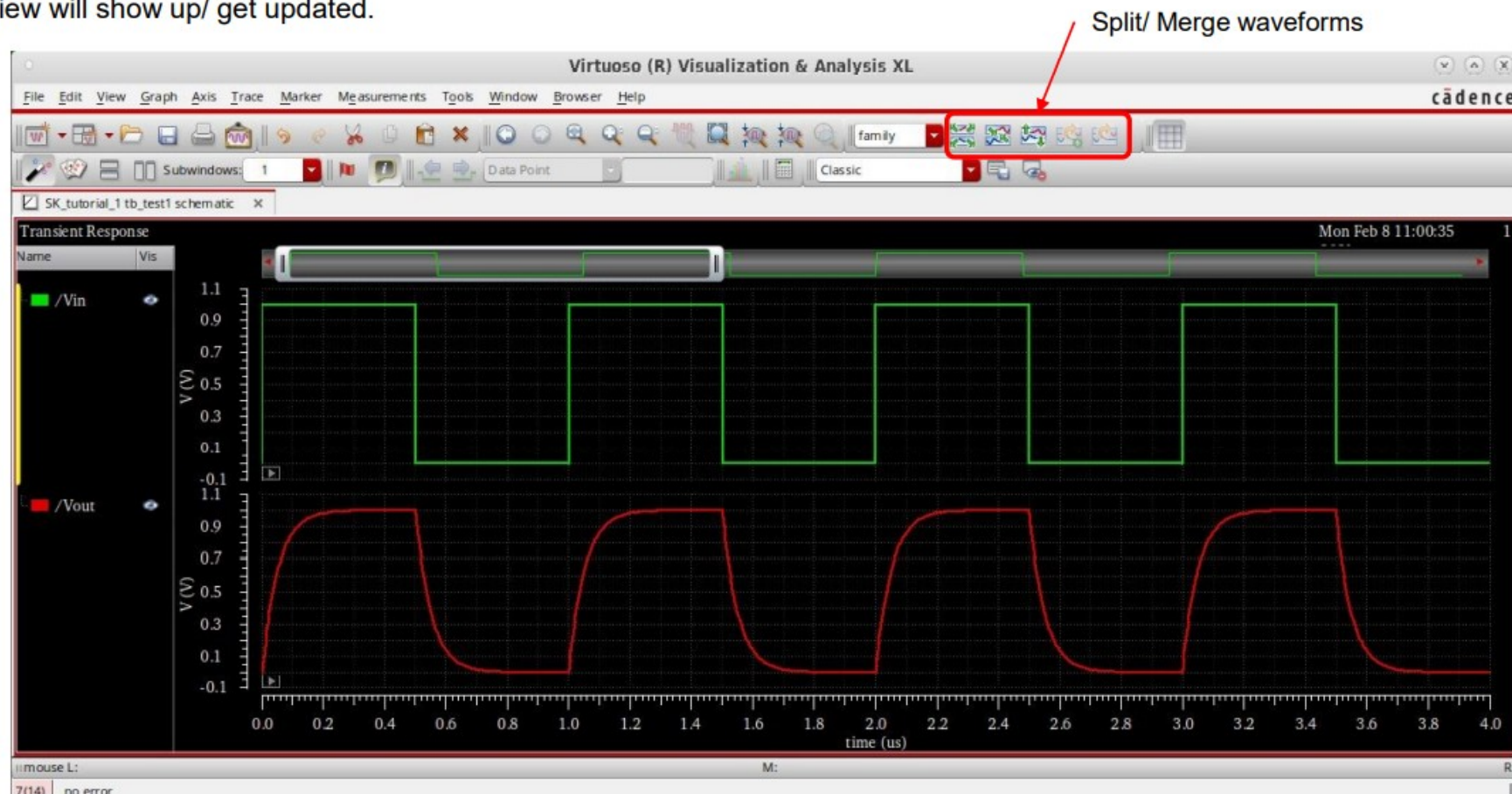
OK Cancel Defaults Apply Help

Plotting a waveform

- To plot a waveform, from ADE: outputs> to be plotted > select on design. Then select the necessary nets. In this case Vin and Vout are selected, which show up in ADE.
- After running the simulation, the waveform view will show up/ get updated.

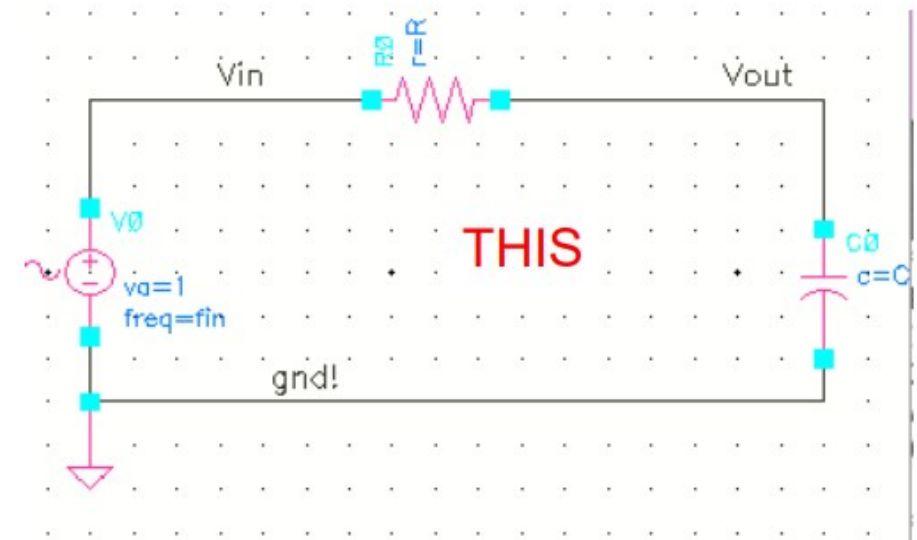
For reports:

- Use medium thickness/ bold lines. Select all the wires (ctrl +A)> rt clk > style > solid/ Width> medium.
- Use the options indication to split and merge the appropriate waveforms together. You can simply drag+drop.
- In order to retain earlier waveforms to compare, say after updating a parameter, you can change the plotting mode in ADE to append instead of replace.
- In order to retain the formatting of the waveforms (line thickness/ color etc..) you should select plotting mode as New window. That way your old plots and formatting is retained. Then, in your old plots, simply ctrl+ R to reload the waveforms with the new simulation.



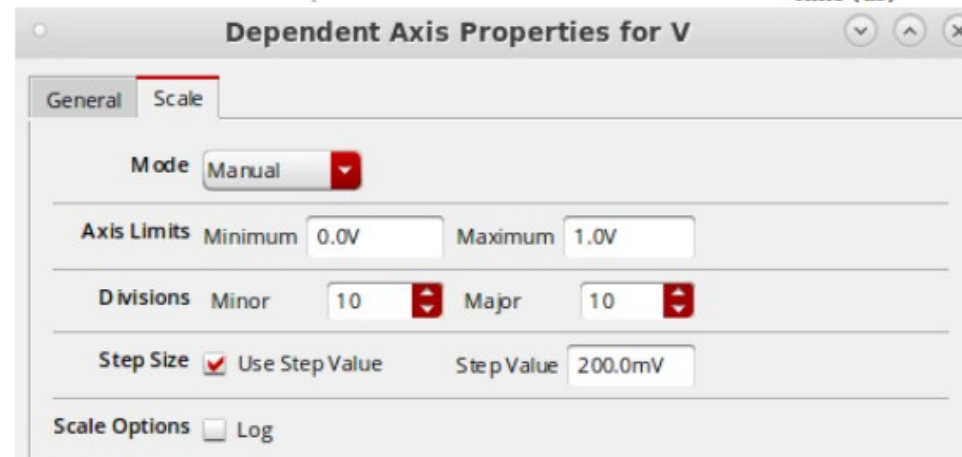
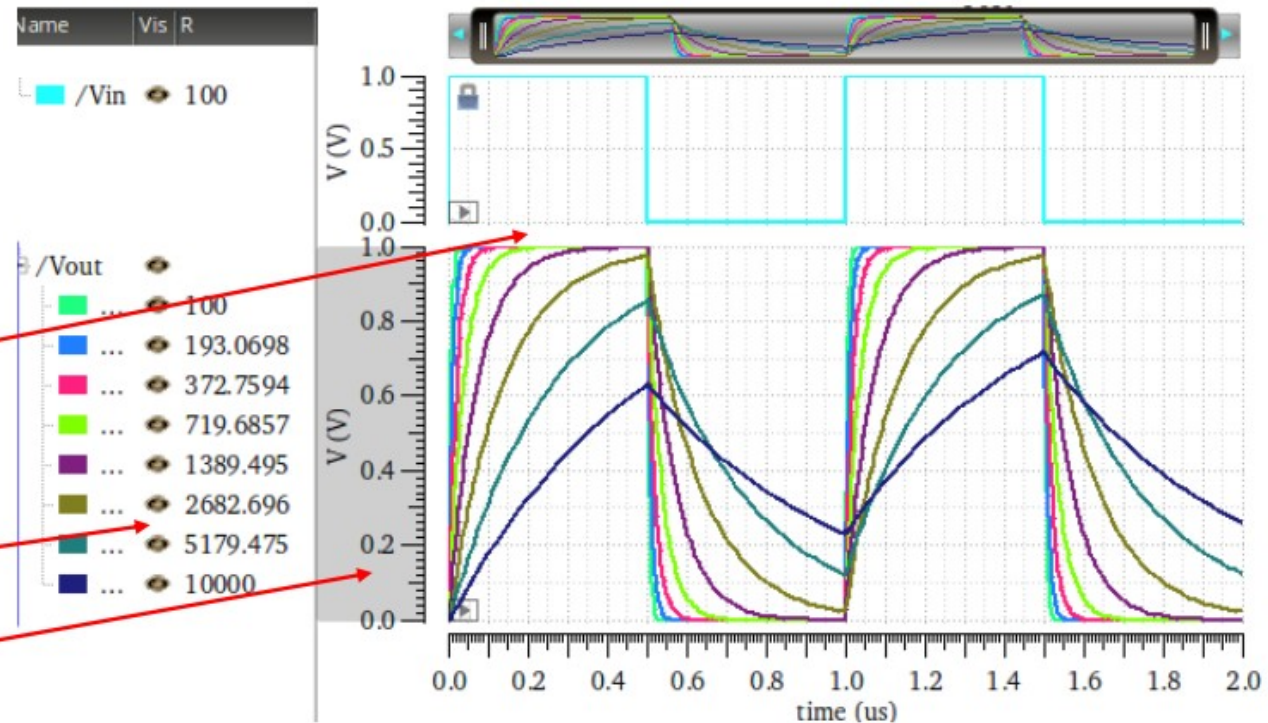
Saving Schematic as image

- Before taking a screenshot of the schematic ensure:
 - All the net highlighting's are removed.
 - The critical device parameters (R,C, W,L, etc.) are visible.
- **Please use a white background.**
- You can simply use your snip tool from windows to take a screenshot. But when doing so make sure you don't have your schematic view maximized.
- Instead size the window approximately to the same aspect ratio of the final image to be inserted into in the report. That way the image quality is drastically improved.
- Use <f>, or **rt +click + drag** to zoom in on only the essential schematic.
- Use snip and save the image as a **.png**, and check for yourself if everything is legible.
- To invert the colors:
 - Open image in paint.
 - Ctrl + A to select the entire image
 - Right click anywhere on image and invert colours



Saving Schematic as image

- A similar procedure can be followed for the waveforms as well.
- Make sure all the lines are medium thickness/ solid.
- Merge and split the necessary waveforms appropriately.
- Size the window to the same size/ aspect ratio of final image.
- For periodic waveforms, there is no point in showing more than 3-4 cycles. So select the time window appropriately.
- Adjust the height of the individual waveform sub-plot as necessary. This can be done by hovering the mouse over the edge and click+drag.
- For parametric analysis/ sweeps make sure the sweep variable is visible.
- Use proper axis scaling. This is best done manually when the waveform window size is reduced. Double click on the axis and go to scale in the pop-up. Select manual and adjust min/ max. select "Use step value" and put in an appropriate number.



Performing a FFT

To get a decent FFT some points are important to remember:

- FFT algorithms work most efficient when using a 2^N number of sample points. Although with current processors this is hardly an issue. For this example $N=16$,

NFFT (no. of sample points) = 65536.

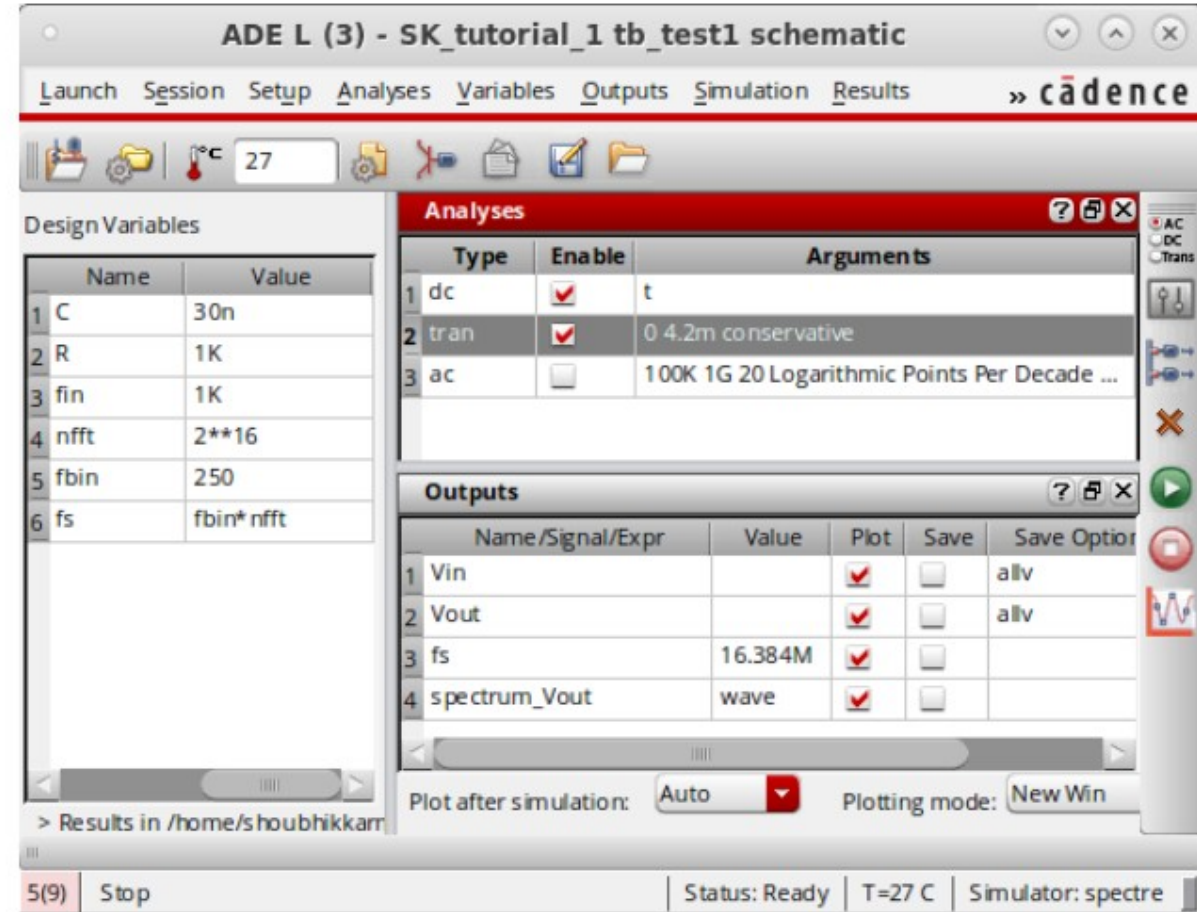
- With a given sample rate f_s , the frequency spectrum has a frequency resolution of

$f_{\text{BIN}} = f_s / \text{NFFT}$. (Also called the resolution bandwidth-RBW).

- To get a decent FFT plot, and avoid signal skirts, it is necessary to ensure that the input frequency satisfies

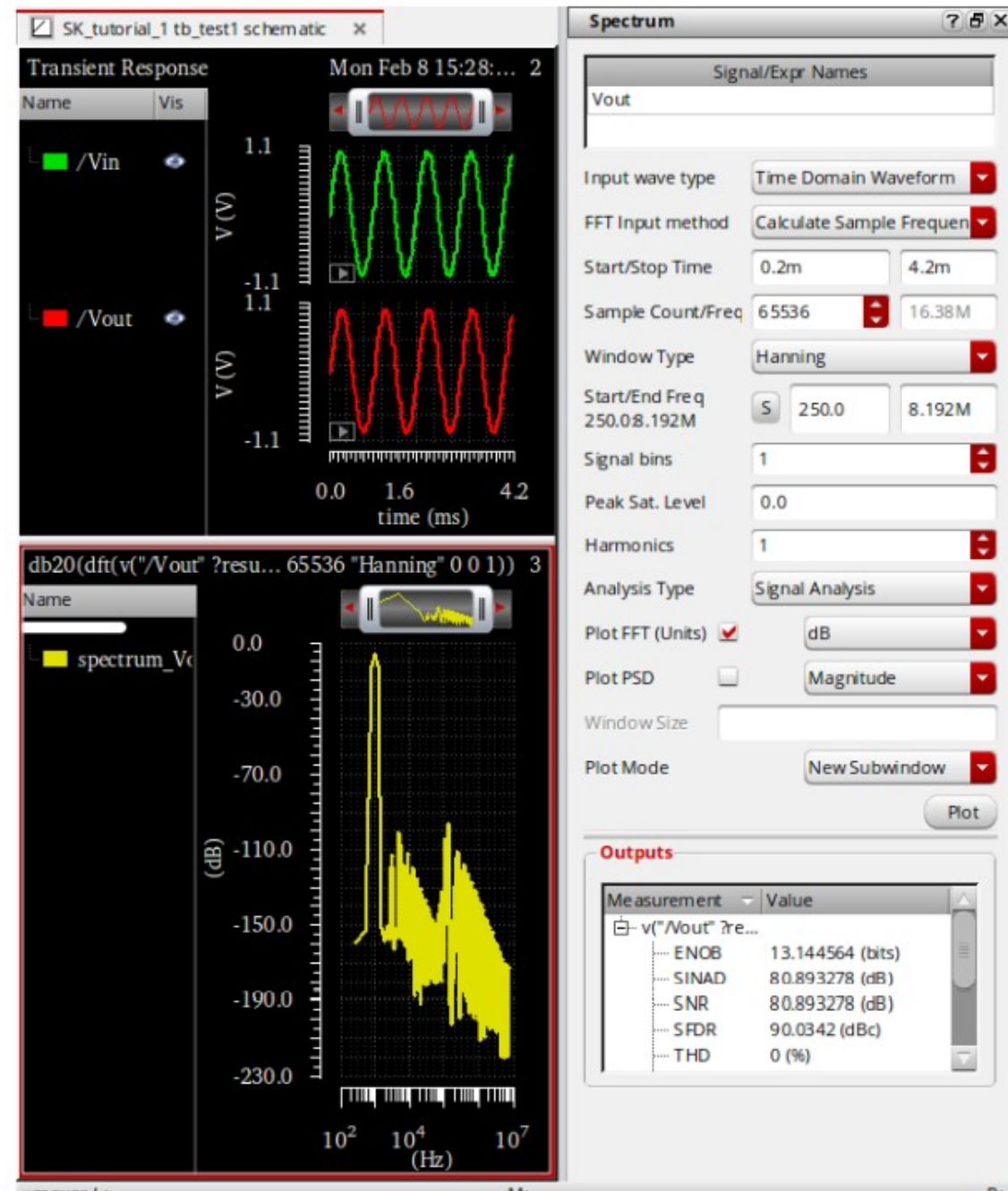
$f_{\text{IN}} = (f_s / \text{NFFT}) * X_{\text{INT}}$, where X_{INT} is an integer. In other words, this is ensuring the input frequency *falls on one of the FFT bins*.

- For this example and f_{BIN} of 250Hz sets **$f_s = 250 * 2^{16} = 16.384\text{MHz}$**
- Total simulation time to get an FFT is **$\text{tsim} = 1/f_{\text{BIN}} = 4\text{ms}$** . It is a good practice to give some additional time to allow for initial transient settling. In this case 200us is given (**$\text{tsim} = 4.2\text{ms}$**)
- These numbers may be inserted as variables in ADE as shown. (Variables > Edit)



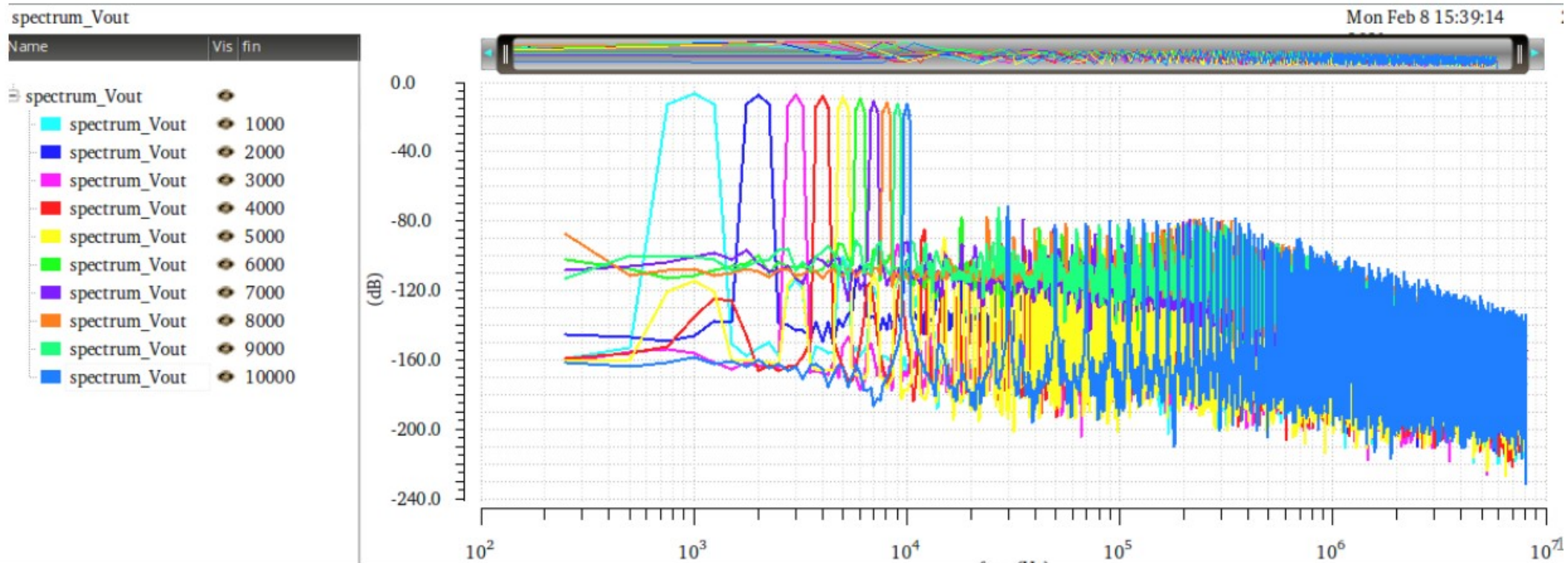
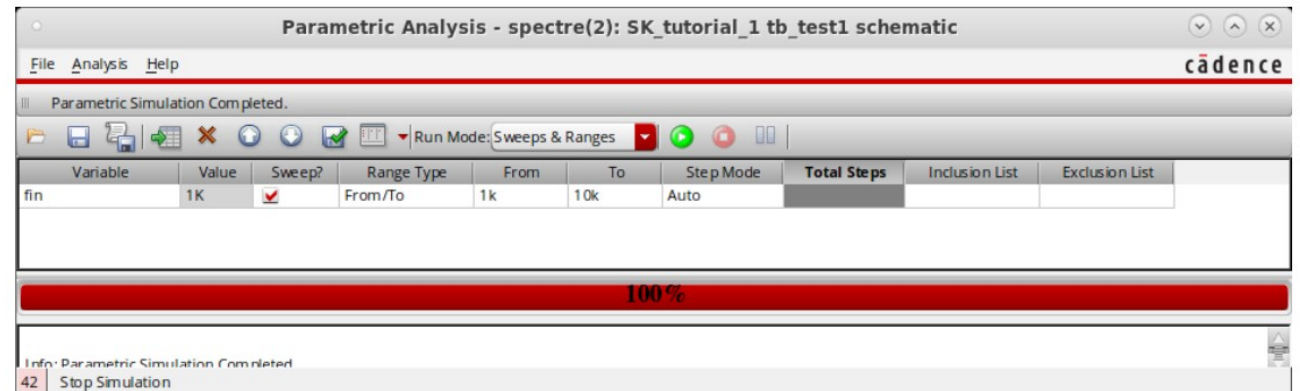
FFTs and Signal-to-Noise-and Distortion (SINAD)

- To plot the FFT, select the waveform, and then Measurements> Spectrum.
- Fill in the values as calculated before. (start/ stop time, Sample count = NFFT)
- Use a Hanning window
- Sync frequencies to auto-calculate fs. If done correctly start frequency is $f_{\text{BIN}} = 250\text{Hz}$, End freq. = $f_s/2 = 8.192\text{MHz}$.
- Use dB and click plot. The FFT should be plotted. Double click X-axis and change it to log. Also change line style to continuous.
- You can rt+click on the plot and Send to> ADE. Next time it gets plotted automatically.
- SINAD can be seen on the right.



Parametric Analysis

- To sweep the input frequency and plot the FFT at one go a parametric sweep may be performed. Tools> Parametric analysis.
- Select fin from the variable list and set up as shown.



Reference

■ Brightspace tutorial1

Tutorial 1 ▼

Upload / Create ▼ Existing Activities ▼

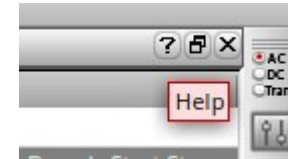
Tutorial-1	PDF document	✓
Tutorial-1 Recording	Video	✓
Server connecting	Word Document	✓

■ Brightspace cadence/spectre tutorial

[cadence/spectre](#) 2

[tutorial](#)

■ Cadence Help



Mini-project

■ Miniproject position

